

Artificial intelligence in animal anatomy: Exploring the technologies, applications, benefits, and challenges

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ABSTRACT

Background: Animal anatomy is revolutionised by use of digital techniques where it is implicated in research, education, and diagnostics. Modern computational tools assist in analysis and mimic human logic and reasoning to solve a problem. Digital technology might assist in handling various issues related to academics and research which encompass ethical issues, unavailability of specimens, and high costs linked with the manual laboratories.

Methods: This review encompasses the application of machine learning (ML) or deep learning (DL) in veterinary anatomical science with focus on computer vision and 3D modelling. ML algorithms are routinely employed to automate medical imaging tasks such as detection, segmentation and classification of structures from images, in a time-efficient and consistent manner compared to traditional methods. The range of applications for modern imaging and visualization tools spans from brain segmentation in preclinical studies to the minor importance of the assessment of highly complex anatomical traits across species. For instance, using micro-computed tomography and sophisticated image processing, researchers are already conducting evolutionary and comparative studies using quick and reliable analysis techniques. In the teaching environment, virtual environments and 3D interactive platforms or models have become useful aids, as students learn better when presented with more engaging and spatially contextualised content. Furthermore, it also addresses the issues and limitations of using cadavers in teaching.

Results: Artificial intelligence (AI) has transformed animal anatomy by improved imaging accuracy, analysis and diagnosis using various algorithms that helped decreasing human error and time. Deep learning-powered augmented reality tools and 3D modelling platforms, such as the Anatomage Table, which enable hands-on training on a digital cadaver, enhance learning with unlimited attempts. Moreover, educational AI models such as *EspINA* and *ZeBraInspector* support volumetric analysis, anatomical reconstruction, and evolutionary research. The diagnosis of liver malignancies, tracheal collapse, and other veterinary conditions has also become more efficient, rapid, and accurate. However, challenges remain, including data scarcity, inconsistencies across species, and limited standardised veterinary datasets comparable to human dataset and frameworks.

Conclusions: AI-driven technologies have revolutionized animal anatomy across various areas including education, research, and diagnosis. They have played a pivotal role in enhancing efficiency, precision, and accessibility while bypassing many ethical hurdles and resource limitations. The combination of machine learning, deep learning, computer vision, virtual modelling, and language models has significantly improved anatomical visualization. Despite these benefits, challenges such as data availability, data heterogeneity, and quality still persist in the use of AI in veterinary anatomy. Future advancements in AI algorithms are expected to increase the precision, efficiency, and impact of veterinary anatomical science.

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1. Introduction

Artificial Intelligence (AI) involves computer-mediated algorithms that can analyze and execute tasks similarly to humans. Artificial intelligence (AI) now influences every aspect of life and academics, and research is no exception. AI represents the intelligence present in machines, which can be trained to process data like the human brain and solve problems (Kour et al., 2022). The term artificial intelligence was first adopted as a field of study in 1956. Recently, there has been much attention drawn towards these technologies due to their use in education, research, and science (Hamet and Tremblay, 2017). Veterinary medicine includes animal health, public health, population health, diseases shared by humans and animals, and production and services (Kour et al., 2022). These technologies have been a great boon in such areas. AI may have the most integral influence on human learning capabilities. The application of AI will have immense influence on human capability to learn (Choudhary et al., 2025). The relevance and use of cadaveric specimens, physical models, and dissections have always been a part of the teaching of veterinary anatomy. Sufficient opportunities to observe, assist, and directly handle pet and farm animals recognise the significance of value of work based learning in professional workplaces. The study conducted on Final-year veterinary students of Chattogram Veterinary and Animal Sciences University (CVASU), Bangladesh to determine the challenges that appeared in work-based learning. It has been observed that work-based learning with more effective learning access is accurate, engaging and interactive and placement for jobs also is achieved. Still, ethical concerns, limited availability of specimens and high costs of necropsy facilities pose challenges (Sattar et al., 2023). The recent shift in anatomical studies is the inclusion of AI with the physical learning in animal anatomical studies, which have been introduced in veterinary curricula. Highly sophisticated three-dimensional (3D) printed models, augmented reality (AR) simulations, and virtual dissection platforms are changing veterinary anatomical studies now. The transition-I aligned with the global requirements, tech-integration incorporated studies in the education (Choudhary and Sarkar, 2025).

Another big challenge faced by students is lack of practical experience due to distance learning (Busch et al., 2024). Using techs like virtual reality and augmented reality makes veterinary teaching less dependent on physical models (Choudhary et al., 2025). Moreover, in veterinary education, incorporating educational games based on AI can enhance understanding through interactive and playful ways (de Brito et al., 2025).

2. Machine learning algorithms

Animal anatomy can be used effectively with computer algorithms to image anatomical structures and recognize anatomical features. According to a report by Brown University, machine learning is being used more and more in medical imaging and eye-tracking. It is being used to annotate gaze patterns, identify physician errors and detect fatigue (Eddine Rahmoun et al., 2025; Ibragimov and Mello-Thoms, 2024). Due to this, it is providing support beyond the diagnostic tools commonly used. At the same time, modern latent machine learning methods are applied to study the cerebral arterial tree, allowing detailed and accurate anatomical analysis. The research utilized multi-network frameworks to successfully identify different cerebral artery types in both stroke patients and healthy participants (Hong et al., 2023). The methodology resulted in less variable outcomes compared to human subjects while achieving superior vessel analysis performance. ML technology has also gained attention in anatomy education. Advances in medical technology now enable users to teach and learn living anatomy through virtual anatomy software and medical imaging (Asad et al., 2023). In 2024, Kohler and colleagues applied machine learning to perform precise brain segmentation in large-scale preclinical imaging. Three different model architectures—U-Net, Attention-U-Net, and DeepLab were trained with over 10,000 magnetic resonance images of rat brains

and compared, which enabled the segmentation of the entire mouse brain into predefined sets. The U-Net outperformed among the three tested. The models trained here not only cancelled out imperfect training data to offer more structured segmentations than registration-based labelling but also performed robustly when presented with input variations. With that the models included uncertainty estimation and an explainability mechanism to provide anomaly detection and quality assurance (Kohler et al., 2024). The increasing use of ML combined with technological resources in anatomy education highlights researchers' growing understanding of these tools' effectiveness for visualizing complex anatomical structures. Applying ML algorithms offers valuable benefits for animal anatomy studies by improving both medical imaging research and educational methods. These applications will continue to grow in effectiveness and scope as technology and datasets (Choudhary et al., 2025).

3. Computer vision and image processing

Multiple image processing and computer vision techniques have been used to advance animal anatomy research through newer analytical methods for studying biological forms and functions. Neuroscience research has used micro-computed tomography (microCT) imaging techniques to analyze the 3D fascicular arrangement of the human vagus nerve. The automation of fascicle segmentation has been performed using neural networks, which has reduced the processing time from almost 40 h to just 24 s (Buyukcelik et al., 2023). The application of computer vision along with image processing techniques to animal anatomy has shed light on their evolutionary development and fine histological details. To capture a high-content 3D image of zebrafish eleuthero-embryos (EEs) at five day post-fertilization (dpf), the ZBI platform was developed. Through the application of a fluorescent lipophilic dye, researchers develop 3D images of embryos in identical orientations for comparison of morphology and other physical properties. The software is useful for performing volumetric analyses of entire EEs and their nervous system white matter (Lempereur et al., 2022).

4. Three-dimensional modeling (3D modelling) and visualization techniques

Three-dimensional modeling and visualization techniques are being effectively utilized in education and research today. These techniques help in elucidating complex anatomical structures. Specimens of pulmonary segments are generated using a 3D printer after software processing, and they are used to educate students alongside traditional methods. Compared to students who received only classical education, the study group was significantly more satisfied with the teaching content and demonstrated better spatial thinking for sectional anatomy (Miao et al., 2023). Studying the central nervous system (CNS) involves understanding how different biological processes influence the brain. EspINA software has been used to analyse magnetic resonance (MR) images to reconstruct 3D images of dog brains that facilitated anatomists to understand the encephalon and the intensity of neurological problems. The tool is highly valuable to understand and teach veterinary neuroanatomy, allowing students to explore the brain in three dimensions and effectively understand the complexity of neuroanatomical structures (Blázquez-Llorca et al., 2023). 3D technology aids in planning treatments in clinical practices. The PIRET (Platform for Treatment Planning in Electroporation-Based Therapies) platform, for example, defines treatment volumes in electroporation-based therapies in dogs to ensure that the electric field reaches the ideal therapeutic threshold. 3D Slicer and modeling software form the basis for its predictions. The 3D modeling techniques and visualization, in recent years, have revolutionized the field of veterinary education and helped students to understand spatial relationships better. Besides, it has improved the quality of instructions and helps to plan accurate treatment. In the veterinary education, integration of advanced tools like three-dimensional (3D)

models, virtual reality (VR), and augmented reality (AR), provides students and faculties a learning atmosphere which is both interactive and dynamic. However issues always surfacing are the data heterogeneity, algorithmic transparency and legal perspectives regarding usage of data. A Balanced approach integrating traditional skills with usage of AI will enhance AI benefits veterinary anatomy and ensure comprehensive veterinary care (Choudhary et al., 2025).

5. Applications of AI in animal anatomy

Recent technological advances have influenced animal anatomy, making research, education, and clinical typing more efficient and precise. Animal models are essential for testing new surgical techniques to determine their efficacy and safety before they are applied to human patients. This minimises associated risks. An animal anatomist's study is also vital to understanding how researchers learn about how organisms respond to surgery, including tissue repair and immune response, which is crucial for understanding surgical outcomes and patient recovery (Choudhary, 2025). Interactive and immersive simulations, powered by AI tools like virtual reality and 3D modeling, have transformed veterinary anatomy education.

5.1. Automated image analysis and segmentation in animal anatomy

Automation and image segmentation have revolutionised the analysis of animal anatomy, especially in medical imaging and preclinical and clinical research. A U-Net convolutional neural network (CNN) was developed to automate fascicle segmentation of human vagus nerves using microCT images. This method has achieved segmentation accuracy four times higher than manual techniques, with a Dice coefficient of 0.87 (den Boers et al., 2023). Understanding disease progression and evaluating treatments require animal models, which are highly valuable. In clinical and preclinical studies, strict control of experimental conditions, baseline measurements before exposure, and longitudinal assessments are maintained, although such control is often lacking in clinical settings. Computed tomography (CT) imaging remains a vital tool for diagnosis, prognosis, and understanding various symptoms of a disease (Vickram et al., 2025). Robotic- assisted minimally invasive oesophagectomy (RAMIE) is employed to improve patient outcomes compared to open surgery. In the study by Li et al. (2024), 27 randomly selected RAMIE recordings were used as a dataset for deep learning applications and compared with AutoLaparo. Among both, the RAMIE dataset is notably more complex, exhibiting richer temporal dynamics and greater variability in phase sequences (den Boers et al., 2023; Li et al., 2024; Tagkalos et al., 2020). Automated deep learning systems integrated with CT imaging have opened new avenues in preclinical and clinical research. For example, in non- human primate (NHP) models of COVID- 19, the entire lung and lesions on CT scans are analysed, achieving Dice coefficients of 99. 4 % for whole- lung segmentation and 60. 2 % for lesion segmentation (Reza et al., 2023). Nonetheless, automation presents challenges, such as the under- detection of smaller fascicles in U- Net CNNs for vagus nerve segmentation, which requires further optimisation (Buyukcelik et al., 2023). Additionally, there is no exact match between NHP anatomy and human anatomy, necessitating species- specific approaches. Overall, advances in imaging techniques have enabled researchers to examine animal anatomy more thoroughly. Ongoing improvements in tissue staining, network architecture, and training data are essential for further progress. A systematic review based on radiography (33 % of the studies), cytology (33 %), health record analysis (8 %), MRI (8 %), environmental data analysis (5 %), photo/video imaging (5 %), and ultrasound (5 %) revealed that DL plays an imperative role in animal disease detection or management (Xiao et al., 2025). The you- only- look- once (YOLO) models, including YOLO v 3, YOLO v 4, and YOLO v 4 tiny, were used by Suksangvoravong et al. (2024) to test canine tracheal collapse using archived lateral cervico- thoracic radiographs. The DL model evaluated cases of grade 1–2 and

grade 3–4 tracheal collapse, achieving a sensitivity of 98. 30 %, a specificity of 99.20 %, and an overall accuracy of 98. 90 % (Suksangvoravong et al., 2024). Here, AI-powered tools may generate a case study for the student to analyse the clinical abnormality. Students can discuss with the chatbot and apply their knowledge in diagnosis. In this way, the thinking and analysis skills of students will be enhanced (Choudhary et al., 2023).

5.2. Virtual dissection and interactive 3D models

Virtual dissection, virtual reality, and interactive 3D models are valuable tools for exploring animal anatomy. In laboratories, photogrammetry-based 3D models of real human brains have been used to teach medical students, leading to higher satisfaction with learning than traditional methods (Aridan et al., 2024). Among the new educational technologies in anatomy is virtual dissection. At the Faculty of Medicine at Al-Balqa University, a software Anatomage table has been prepared to enable students to perform hands-on manipulation of a digital cadaver via an electronic screen in the form of a table. It is a high-tech tool that allows students to perform hands-on manipulation of a digital cadaver via an electronic screen in the form of a table. The study was conducted to evaluate Jordanian medical students' attitudes towards virtual dissection technology as part of their learning experience, aiming to determine how effective a virtual dissection tool can be. With a total of 414 participants, at least 50 % of the participants found improvements in their grades and almost 64 % reported that the tool helped them memorise anatomical details more effectively (Funjan et al., 2023). A schematic depiction of horse dissection, including the uterus, placenta, and ovary sectioned to observe ovulation stages, is presented in Fig. 1.

Standardised high-quality training data is a prerequisite for surgical and anatomical studies. The veterinary subject inherently encompasses a diverse range of species; thus, fostering national/international collaboration is not only beneficial but also mandatory to improve data quality and fine-tune AI models. The formation of an international consortium focused on veterinary phenomics for AI is suggested (Xiao et al., 2025).

5.3. Comparative anatomy across species

A comparative analysis of species-specific anatomy is also essential to provide valuable insights for evolutionary and developmental studies. Studies on neuroanatomy in various animal models have significantly improved knowledge of how species are alike and different. Dog functional magnetic resonance imaging (fMRI) is commonly employed to explore brain functions and evolutionary development. Image acquisition during dog MRI is performed using coils similar to those used in human MRI; however, these setups are designed for humans and need to be optimized for canine use. A multichannel receive coil has been developed that enables high-resolution functional imaging and reveals canine cranial anatomy (Guran et al., 2023).

5.4. Anomaly detection and disease diagnosis

Computational approaches have performed very well in diagnosing diseases and are now widely used for detecting anomalies and disease conditions in veterinary medicines (Xiao et al., 2025). Integration of artificial intelligence (AI) and machine learning (ML) technologies, offers a transformative step toward enhancing early identification and tailored interventions where various parameters like serum biomarkers, phenotypic traits and medical images may be employed for improved diagnosis with high accuracy (Koul et al., 2023). Different machine learning approaches can be used to model diseases that include random forest (RF), boosted regression tree (BRT), artificial neural network (ANN), multiple adaptive regression spline (MARS), flexible discriminant analysis (FDA) and classification tree analysis (CTA). Using a risk

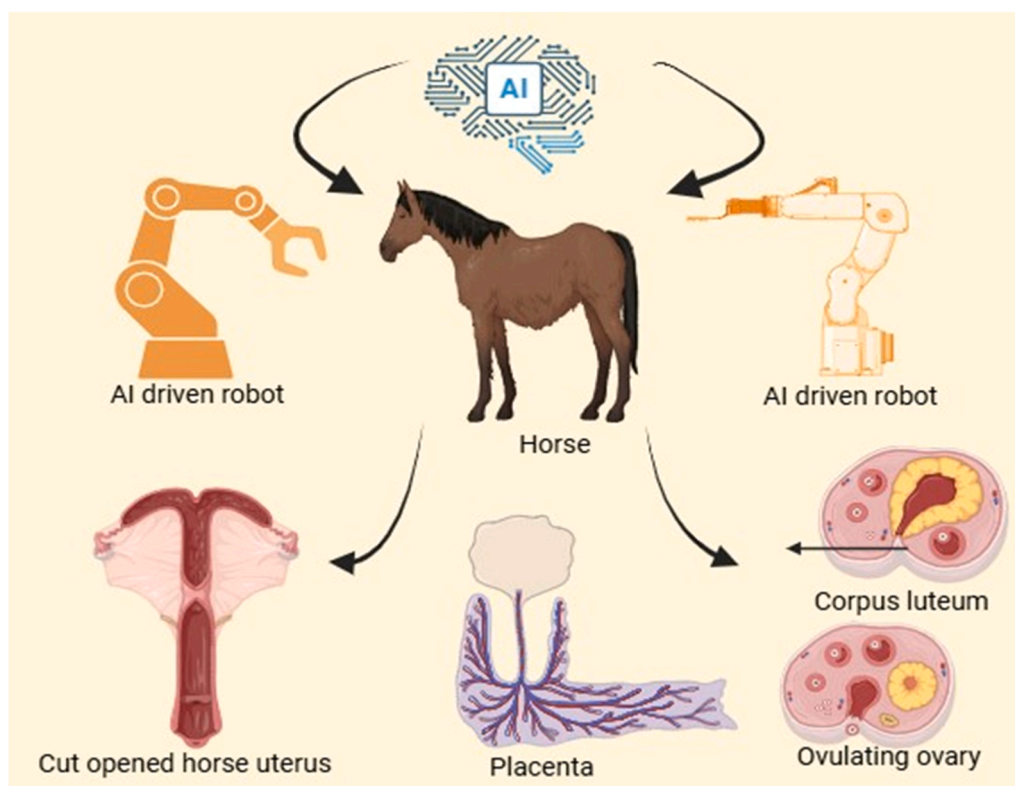


Fig. 1. Schematic illustration of AI-driven robotic dissection of a horse.

model, animals can be classified into a hierarchical scale of risk from low to high. Li et al. (2020) used a computational approach to detect left atrial enlargement on canine thoracic radiographs. The CNN model which was trained, validated, and tested using 792 radiographs achieved an accuracy, sensitivity, and specificity of 82.71 %, 68.42 %, and 87.09 %, respectively (Li et al., 2020). A schematic illustration is given in Fig. 2. Horses suffering from acute abdominal colic (colic) requiring surgery can be predicted using two models developed by Fraiwan and

Abutarbush (2020) (Fraiwan and Abutarbush, 2020), which estimate the likelihood of survivability based on clinical data. They used Decision Trees, Multilayer Perceptron, Bayes Network, and Naïve Bayes algorithms, with Naïve Bayes achieving the highest precision of 97.4 %. The models predicted that nearly 10 % of colic cases would require surgical intervention. Canine hypoadrenocorticism (CHA) affects roughly 0.3 % of dogs and is usually life-threatening, involving the kidneys and gastrointestinal system. Since the disease is fatal, early detection is

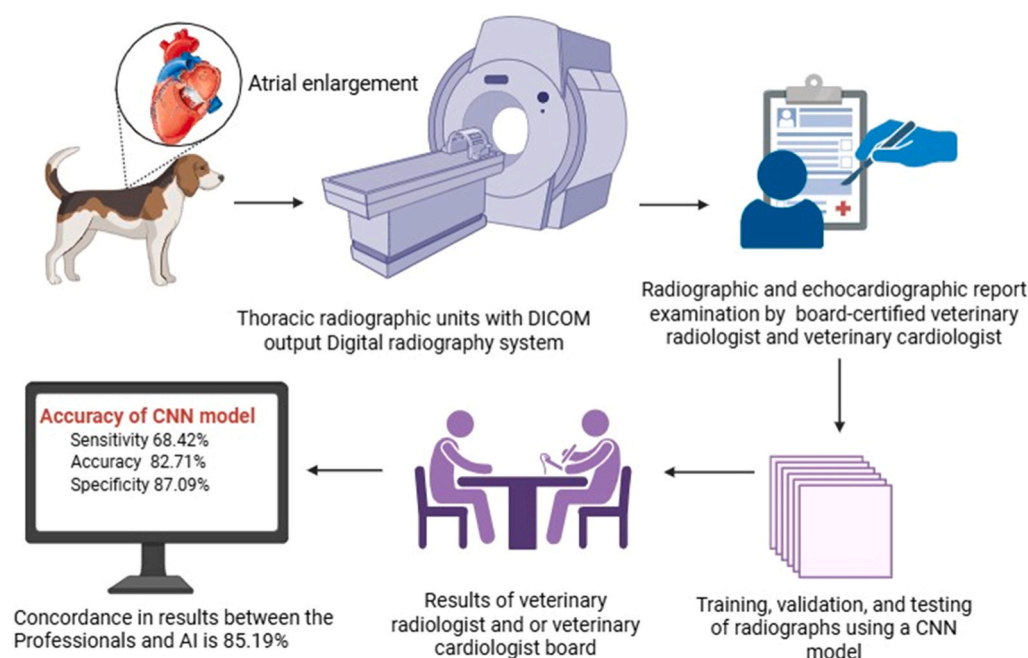


Fig. 2. Schematic illustration of the work by Li et al. (2020) (Li et al., 2020), showing an overall agreement of 85.19 % between AI and veterinary professionals.

crucial. However, lack of awareness and low suspicion rates pose challenges. Reagan et al. (2022) evaluated 908 control dogs alongside 133 dogs diagnosed with the disease. The model was trained on 80 % of the data and tested on the remaining portion, achieving a sensitivity of 96.3 % and a specificity of 97.2 % (Reagan et al., 2022).

Subclinical bovine mastitis negatively affects milk production and quality, and excessive use of antibiotics raises the risk of bacterial resistance. For prediction, data from 364,249 milking instances were collected, including milk volume, lactose concentration, electrical conductivity, protein concentration, peak flow, and milking time. Various prediction methods were tested, with the Generalized Linear Model showing high precision and accuracy at 84.9 % (Ebrahimi et al., 2019). In veterinary medicine, feline reticulocyte counting is still performed manually. Eight hundred labelled images were used with open-source Keras, achieving 98.7 % prediction accuracy in detecting aggregated feline reticulocytes in blood smear slides, surpassing human performance (Vinicki et al., 2018).

Tumour heterogeneity is a vital characteristic of all highly aggressive neoplastic processes and is linked to local recurrence and distant metastasis. Quantitative analysis of CT textural features could serve as an indirect method to predict tumour progression. A study by Shaker and colleagues developed a predictive model for malignancy using 40 dogs with confirmed liver tumours, employing triphasic image acquisition to quantitatively evaluate CT heterogeneity in dogs. The results demonstrated that postcontrast CT hepatic mass contours were found to be more informative and held clinical utility as a non-invasive method for predicting hepatic malignancy (Shaker et al., 2021). A deep convolutional neural network (DNN) was developed to identify degenerative hepatic disease in dogs using ultrasound images of the liver and was compared with the diagnostic accuracy of serum biochemistry and cytology. A total of 52 subjects were included with no prior history neoplasia, or ascites. Using transfer learning, the pretrained neural network was retrained to finely tune the data set, resulting in higher diagnostic accuracy compared to conventional cytology and serum biochemical markers (alanine transaminase and aspartate transaminase) based assays with 100 % sensitivity and 82.8 % specificity (Banzato et al., 2018).

6. Challenges and limitations

Like other scientific disciplines, animal anatomy faces challenges related to data quality and accessibility. The lack of a reliable reporting system that relies on large datasets makes data collection difficult. Data is often scattered and inconsistent in format, hindering research and analysis. Since detailed studies require labelled data, manually annotating large volumes presents a major challenge (Anwar et al., 2018). Achieving high accuracy and precision demands close collaboration between IT specialists and veterinarians. Electronic records and data from health centres, research labs, and veterinary institutions must collaborate (Esteve et al., 2019). Similar to human healthcare systems, where data is systematically collected and well-annotated following the guidelines of the Systematized Nomenclature of Medicine Clinical Terms (SNOMED-CT), the Small Animal Veterinary Surveillance Network (SAVSNET) has been established. A significant challenge in data retrieval is that, because data is often freely accessible, obtaining answers to research questions can be difficult and heavily reliant on data sharing willingness. The consistency of data collection greatly affects results. Both data quality and quantity influence AI outputs. In animal anatomy, this challenge is especially acute because veterinary medicine involves many different animals, and variations in their anatomies and physiologies complicate the development of universal AI models. Large language models (LLMs) are a subset of generative AI capable of human-like text generation. LLMs consist of millions of parameters and are pretrained on extensive textual datasets. Recently, veterinary writing styles have shifted from general internet content, using different vocabulary and acronyms. Most LLMs use pretraining corpora from the

internet. Jiang et al. (2024) developed VetLLM, designed to extract diagnostic details from veterinary notes, and it demonstrated reasonable performance even when encountering cases it had not previously seen. A newer AI method, DeepTag, uses training data from the Colorado State University Veterinary (CSU) dataset; however, its performance drops sharply when notes from private practices are used (Jiang et al., 2024). VetTag is another LLM that outperforms DeepTag by predicting 4, 4577 veterinary diagnosis codes from free text (Zhang et al., 2019).

7. Conclusion and future perspectives

Artificial intelligence (AI) and related technologies are beginning to transform veterinary anatomy, influencing research, education, diagnostics, and clinical practice. The use of machine learning (ML), deep learning (DL), computer vision, and advanced 3D visualization tools has opened new opportunities for understanding animal structure and function. Animals are being taught anatomy as they experience a major shift, assisted by ML algorithms. Manual methods may show considerable variability in results; software-based approaches offer greater consistency, speed, precision, and other benefits. Well-designed, intelligent medical software can alert clinicians when fatigue or errors might affect diagnoses in imaging. It not only provides essential diagnostic support but can also elevate the study of animal anatomy. Computer vision and image processing techniques, including microCT, high-content 3D imaging, and neural network-driven segmentation, along with platforms like ZeBraInspector (ZBI), enable both high-throughput and volumetric quantification in comparative anatomy. Studies have shown that combining traditional methods with AI models enhances student performance in anatomy education. Furthermore, AI has significantly influenced the development of animal anatomy curricula.

Future trends are likely to drive the next stage of development in animal anatomy. Sharing data to build detailed multispecies anatomical datasets through collaboration will be advantageous. Techniques such as semi-supervised, unsupervised, and transfer learning can assist in managing large datasets and enable highly customised diagnosis, prognosis, and treatment. Moving forward, approaches that improve transparency and interpretability will be essential for fostering trust in research and clinical environments, allowing veterinarians to assess and confirm decisions. The latest advances in animal anatomy herald a new era of accuracy, efficiency, and discovery. With continuous technological progress, interdisciplinary collaboration, meticulous methodology, and ethical oversight, expertise, veterinary care, and knowledge of animal biology will continue to evolve.

CRediT authorship contribution statement

Rekha Khandia: Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization. **Pankaj Gurjar:** Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Somya Rajput:** Writing – original draft, Visualization, Methodology, Investigation.

Ethics and consent to participate declarations

Not applicable.

Clinical trial number

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Declaration of Competing Interest

The authors declare that they have no known competing financial or personal interests that could have appeared to influence the work reported in this manuscript.

References

- Anwar, S.M., Majid, M., Qayyum, A., Awais, M., Alnowami, M., Khan, M.K., 2018. Medical image analysis using convolutional neural networks: a review. *J. Med. Syst.* 42, 226. <https://doi.org/10.1007/s10916-018-1088-1>.
- Aridan, N., Bernstein-Eliav, M., Gamzo, D., Schmeidler, M., Tik, N., Tavor, I., 2024. Neuroanatomy in virtual reality: development and pedagogical evaluation of photogrammetry-based 3D brain models. *Anat. Sci. Educ.* 17, 239–248. <https://doi.org/10.1002/ase.2359>.
- Asad, M.R., Al Mutairi, A., AlZahrani, R.E., Ahmed, M.M., Nazeer, M., Taha, M., 2023. Role of living anatomy in medical education: a narrative review. *J. Pharm. Bioallied Sci.* 15, S843–S845. <https://doi.org/10.4103/jpbs.jpbs.235.23>.
- Banzato, T., Bonsembiante, F., Aresu, L., Gelain, M.E., Burti, S., Zotti, A., 2018. Use of transfer learning to detect diffuse degenerative hepatic diseases from ultrasound images in dogs: a methodological study. *Vet. J. Lond. Engl.* 1997 (233), 35–40. <https://doi.org/10.1016/j.tvjl.2017.12.026>.
- Blázquez-Llorca, L., Morales de Paz, L., Martín-Orti, R., Santos-Álvarez, I., Fernández-Valle, M.E., Castejón, D., García-Real, M.I., Salguero-Fernández, R., Pérez-Lloret, P., Moreno, N., Jiménez, S., Herrero-Fernández, M.J., González-Soriano, J., 2023. The application of 3D anatomy for teaching veterinary clinical neurology. *Anim. Open Access J. MDPI* 13, 1601. <https://doi.org/10.3390/ani13101601>.
- den Boer, R.B., Jaspers, T.J.M., de Jongh, C., Pluim, J.P.W., van der Sommen, F., Boers, T., van Hillegersberg, R., Van Eijnatten, M. a J.M., Ruurda, J.P., 2023. Deep learning-based recognition of key anatomical structures during robot-assisted minimally invasive esophagectomy. *Surg. Endosc.* 37, 5164–5175. <https://doi.org/10.1007/s00464-023-09990-z>.
- de Brito, C., Redondo, J.I., Tadeo-Cervera, I., Bataller, E., García-Roselló, M., Cuquerella Madoz, I., Terrado, J., 2025. Integrating artificial intelligence into veterinary education: student perspectives. *Front. Vet. Sci.* 12, 1641685. <https://doi.org/10.3389/fvets.2025.1641685>.
- Busch, F., Hoffmann, L., Truhn, D., Ortiz-Prado, E., Makowski, M.R., Bressemer, K.K., Adams, L.C., COMFORT Consortium, 2024. Global cross-sectional student survey on AI in medical, dental, and veterinary education and practice at 192 faculties. *BMC Med. Educ.* 24, 1066. <https://doi.org/10.1186/s12909-024-06035-4>.
- Buyukcelik, O.N., Lapierre-Landry, M., Kolluru, C., Upadhye, A.R., Marshall, D.P., Pelot, N.A., Ludwig, K.A., Gustafson, K.J., Wilson, D.L., Jenkins, M.W., Shoffstall, A. J., 2023. Deep-learning segmentation of fascicles from microCT of the human vagus nerve. *Front. Neurosci.* 17, 1169187. <https://doi.org/10.3389/fnins.2023.1169187>.
- Choudhary, O.P., 2025. Animal models for surgeries and implants: a vital tool in medical research and development. *Ann. Med. Surg.* 12 (87), 4090–4095. <https://doi.org/10.1097/MS9.00000000000003400>.
- Choudhary, O.P., Sarkar, R., 2025. Animal anatomical teaching models for enhanced veterinary anatomy education and learning. *Pak. Vet. J.* 45, 961. <https://doi.org/10.29261/pakvetj/2025.250>.
- Choudhary, O.P., Saini, J., Challana, A., 2023. ChatGPT for veterinary anatomy education: an overview of the prospects and drawbacks. *Int. J. Morphol.* 41, 1198–1202. <https://doi.org/10.4067/S0717-95022023000401198>.
- Choudhary, O.P., Infant, S.S., As, V., Chopra, H., Manuta, N., 2025. Exploring the potential and limitations of artificial intelligence in animal anatomy. *Ann. Anat. Anat. Anz. Off. Organ Anat. Ges.* 258, 152366. <https://doi.org/10.1016/j.aanat.2024.152366>.
- Ebrahimi, M., Mohammadi-Dehcheshmeh, M., Ebrahimi, E., Petrovski, K.R., 2019. Comprehensive analysis of machine learning models for prediction of sub-clinical mastitis: deep learning and gradient-boosted trees outperform other models. *Comput. Biol. Med.* 114, 103456. <https://doi.org/10.1016/j.compbimed.2019.103456>.
- Eddine Rahmoun, D., Abdel Maksoud, M.K.M., Lieschova, M., Mylostyvyi, R., 2025. Applications of artificial intelligence in veterinary anatomical pathology: enhancing diagnosis and treatment in animal healthcare. *Acta Vet. Eurasia* 51, 1–7. <https://doi.org/10.5152/actavet.2025.25068>.
- Esteva, A., Robicquet, A., Ramsundar, B., Kuleshov, V., DePristo, M., Chou, K., Cui, C., Corrado, G., Thrun, S., Dean, J., 2019. A guide to deep learning in healthcare. *Nat. Med.* 25, 24–29. <https://doi.org/10.1038/s41591-018-0316-z>.
- Fraiman, M.A., Abutarbush, S.M., 2020. Using artificial intelligence to predict survivability likelihood and need for surgery in horses presented with acute abdomen (Colic). *J. Equine Vet. Sci.* 90, 102973. <https://doi.org/10.1016/j.jevs.2020.102973>.
- Funjan, K., Ashour, L., Salameh, M., Mustafa, A., Seed Ahmed, M., 2023. Perceptions and attitudes of Jordanian medical students on using 3D interactive anatomy dissection in teaching and learning anatomy. *Adv. Med. Educ. Pract.* 14, 837–844. <https://doi.org/10.2147/AMEP.S419333>.
- Guran, C.-N.A., Sladky, R., Karl, S., Boch, M., Laistler, E., Windischberger, C., Huber, L., Lamm, C., 2023. Validation of a new coil array tailored for dog functional magnetic resonance imaging studies. *eNeuro* 10, ENEURO.0083-22.2022. <https://doi.org/10.1523/ENEURO.0083-22.2022>.
- Hamet, P., Tremblay, J., 2017. Artificial intelligence in medicine. *Metabolism* 69S, S36–S40. <https://doi.org/10.1016/j.metabol.2017.01.011>.
- Hong, S.-W., Song, H.-N., Choi, J.-U., Cho, H.-H., Baek, I.-Y., Lee, J.-E., Kim, Y.-C., Chung, D., Chung, J.-W., Bang, O.-Y., Kim, G.-M., Park, H.-J., Liebeskind, D.S., Seo, W.-K., 2023. Automated in-depth cerebral arterial labelling using cerebrovascular vasculature reframing and deep neural networks. *Sci. Rep.* 13, 3255. <https://doi.org/10.1038/s41598-023-30234-6>.
- Ibragimov, B., Mello-Thoms, C., 2024. The use of machine learning in eye tracking studies in medical imaging: a review. *IEEE J. Biomed. Health Inf.* 28, 3597–3612. <https://doi.org/10.1109/JBHI.2024.3371893>.
- Jiang, Y., Irvin, J.A., Ng, A.Y., Zou, J., 2024. VetLLM: large language model for predicting diagnosis from veterinary notes. *Pac. Symp. Biocomput. Pac. Symp. Biocomput.* 29, 120–133.
- Kohler, J., Bielser, T., Adaszewski, S., Künnecke, B., Bruns, A., 2024. Deep learning applied to the segmentation of rodent brain MRI data outperforms noisy ground truth on full-fledged brain atlases. *NeuroImage* 304, 120934. <https://doi.org/10.1016/j.neuroimage.2024.120934>.
- Koul, A.M., Ahmad, F., Bhat, A., Aein, Q.-U., Ahmad, A., Reshi, A.A., Kaul, R.-U.-R., 2023. Unraveling down syndrome: from genetic anomaly to artificial intelligence-enhanced diagnosis. *Biomedicine* 11, 3284. <https://doi.org/10.3390/biomedicine11123284>.
- Kour, savleen, Agrawal, R., Sharma, N., Tikoo, A., Pande, N., Sawhney, A., 2022. Artificial intelligence and its application in animal disease diagnosis. *J. Anim. Res.* 12. <https://doi.org/10.30954/2277-940X.01.2022.1>.
- Lempereur, S., Machado, E., Licata, F., Simion, M., Buzer, L., Robineau, I., Hémond, J., Banerjee, P., De Crozé, N., Léonard, M., Affaticati, P., Talbot, H., Joly, J.-S., 2022. ZeBrainInspector, a platform for the automated segmentation and analysis of body and brain volumes in whole 5 days post-fertilization zebrafish following simultaneous visualization with identical orientations. *Dev. Biol.* 490, 86–99. <https://doi.org/10.1016/j.ydbio.2022.07.004>.
- Li, S., Wang, Z., Visser, L.C., Wisner, E.R., Cheng, H., 2020. Pilot study: application of artificial intelligence for detecting left atrial enlargement on canine thoracic radiographs. *Vet. Radiol. Ultrasound Off. J. Am. Coll. Vet. Radiol. Int. Vet. Radiol. Assoc.* 61, 611–618. <https://doi.org/10.1111/vru.12901>.
- Li, Y., Jaarsveld, R. van, Jong, R. de, Bongers, J., Kuiper, G., Hillegersberg, R. van, Ruurda, J., Breeuwer, M., Khalil, Y.A., 2024. Benchmarking and Enhancing Surgical Phase Recognition Models for Robotic-Assisted Esophagectomy. <https://doi.org/10.48550/arXiv.2412.04039>.
- Miao, H., Ding, J., Gong, X., Zhao, J., Li, H., Xiong, G., Zhou, X., Liu, W., Wu, F., 2023. Application of 3D-printed pulmonary segment specimens in experimental teaching of sectional anatomy. *BMC Surg.* 23, 109. <https://doi.org/10.1186/s12893-023-02022-6>.
- Reagan, K.L., Pires, J., Quach, N., Gilor, C., 2022. Evaluation of a machine learning tool to screen for hypoadrenocorticism in dogs presenting to a teaching hospital. *J. Vet. Intern. Med.* 36, 1942–1946. <https://doi.org/10.1111/jvim.16566>.
- Reza, S.M.S., Chu, Winston T., Homayounieh, F., Blain, M., Firoozabadi, F.D., Anari, P.Y., Lee, J.H., Worwa, G., Finch, C.L., Kuhn, J.H., Malayeri, A., Crozier, I., Wood, B.J., Feuerstein, I.M., Solomon, J., 2023. Deep-learning-based whole-lung and lung-lesion quantification despite inconsistent ground truth: application to computerized tomography in SARS-CoV-2 nonhuman primate models. *Acad. Radio.* 30, 2037–2045. <https://doi.org/10.1016/j.acra.2023.02.027>.
- Sattar, A.A., Hoque, M.A., Irin, N., Charles, D., Ciappesoni, J.L., Anwer, M.S., Debnath, N., Baillie, S., 2023. Identifying benefits, challenges, and options for improvement of veterinary work-based learning in Bangladesh. *J. Vet. Med. Educ.* 50, 570–581. <https://doi.org/10.3138/jvme-2022-0049>.
- Shaker, R., Wilke, C., Ober, C., Lawrence, J., 2021. Machine learning model development for quantitative analysis of CT heterogeneity in canine hepatic masses may predict histologic malignancy. *Vet. Radiol. Ultrasound Off. J. Am. Coll. Vet. Radiol. Int. Vet. Radiol. Assoc.* 62, 711–719. <https://doi.org/10.1111/vru.13012>.
- Suksangvoravong, H., Choinsunirachon, N., Tongloy, T., Chuwongin, S., Boonsang, S., Kittichai, V., Thanaboonipat, C., 2024. Automatic classification and grading of canine tracheal collapse on thoracic radiographs by using deep learning. *Vet. Radiol. Ultrasound Off. J. Am. Coll. Vet. Radiol. Int. Vet. Radiol. Assoc.* 65, 679–688. <https://doi.org/10.1111/vru.13413>.
- Tagkalos, E., Goense, L., Hoppe-Lotichius, M., Ruurda, J.P., Babic, B., Hadzijušević, E., Kneist, W., van der Sluis, P.C., Lang, H., van Hillegersberg, R., Grimmering, P.P., 2020. Robot-assisted minimally invasive esophagectomy (RAMIE) compared to conventional minimally invasive esophagectomy (MIE) for esophageal cancer: a propensity-matched analysis. *Dis. Esophagus Off. J. Int. Soc. Dis. Esophagus* 33, doz060. <https://doi.org/10.1093/dote/doz060>.
- Vickram, A.S., Infant, S.S., Priyanka, null, Chopra, H., 2025. AI-powered techniques in anatomical imaging: impacts on veterinary diagnostics and surgery. *Ann. Anat. Anat. Anz. Off. Organ Anat. Ges.* 258, 152355. <https://doi.org/10.1016/j.aanat.2024.152355>.
- Vinicki, K., Ferrari, P., Belic, M., Turk, R., 2018. Using Convolutional Neural Networks for Determining Reticulocyte Percentage in Cats. <https://doi.org/10.48550/arXiv.1803.04873>.
- Xiao, S., Dhand, N.K., Wang, Z., Hu, K., Thomson, P.C., House, J.K., Khatkar, M.S., 2025. Review of applications of deep learning in veterinary diagnostics and animal health. *Front. Vet. Sci.* 12, 1511522. <https://doi.org/10.3389/fvets.2025.1511522>.
- Zhang, Y., Nie, A., Zehnder, A., Page, R.L., Zou, J., 2019. VetTag: improving automated veterinary diagnosis coding via large-scale language modeling. *NPJ Digit. Med.* 2, 35. <https://doi.org/10.1038/s41746-019-0113-1>.